

7. GOVERNANCE, LINEAGE & RELEASE GATE

STPA Hazard Mitigation · Bumpless Human Joystick Override · Defensible Release Case (**LOOP-01** → **REL-01**)

1. SENSING

Photons / Voltages to DMA
MIPI CSI-2 · I2C · SPI Bus Priority

2. PERCEPTION

Encoders to Spatial Tokens
ViT · DINOv2 · 3D Affordances

3. WORLD MODELS

Latent Belief & SE(3) Trees
JEPA / RSSM Dynamics · TTL Validity

5. TRAJECTORY DECODERS

Diffusion Policies · ACT Action Chunking (H -steps) · Jerk Bounds

4. SEMANTIC DELIBERATION

VLMs & Spatial Foundation Models · Goals · Leases (t_{expire})

Expiring Proposal p_t

THE PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (NO DIRECT MOTOR ACCESS)

6. REAL-TIME REFLEX & SAFETY ENFORCEMENT

1 kHz Reflex Timing Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}}(v_t) \leq d_{\text{clearance}}$ · Veto (u_t)

Permitted Action $u_t = \text{permit}(p_t)$

THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)

Kinetic Momentum ($p=mv$) · Joule Heat Dissipation (I^2R) · Matter Mutation · Friction (μ) · Collision Dynamics

Endogenous Sensory Shift ($A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$)